Yash Vijay Tupe

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Professional Summary

Aspiring AI/ML researcher and data scientist deeply passionate about advancing applied research in artificial intelligence, machine learning, and generative AI for healthcare and emerging technologies. Proven ability to design, build, and deploy end-to-end ML and LLM-integrated systems achieving up to 92% accuracy using Python, LangChain, and Flask. Motivated by interdisciplinary R&D challenges at the intersection of deep learning, healthcare-tech, and smart-city infrastructure, and eager to contribute rigorous, hands-on technical work to a collaborative research environment.

Education

P.V.G. College of Engineering & S.S. Dhamankar Institute of Technology, Nashik **June 2024 – June 2027**
Bachelor of Engineering (B.E.), Information Technology — Affiliated to Savitribai Phule Pune University CGPA: 8.5 / 10

GES's Sir Dr. M. S. Gosavi Polytechnic, Nashik **July 2021 – June 2024**
Diploma in Computer Engineering Percentage: 85.71%

Skills

Programming Languages	Python, Java, SQL
Machine Learning & AI	Scikit-learn, Random Forest, Deep Learning, CNNs, Supervised Learning, Feature Engineering, Model Evaluation, Model Deployment
LLM & Generative AI	LangChain, OpenAI API, Prompt Engineering, RAG, LLM Integration
Data Science	Pandas, NumPy, EDA, Data Preprocessing, Statistical Analysis, Matplotlib, Seaborn
Frameworks & Tools	Flask, Firebase, REST API, SQL, Git, GitHub, Jupyter Notebook, VS Code
Deployment & MLOps	Docker, AWS, Render, Version Control, CI/CD Basics
Soft Skills	Cross-functional Collaboration, Problem-Solving, Communication, Adaptability, Time Management

Work Experience

AICTE x IBM x Edunet Foundation **Jan – Feb 2026**
AI & Machine Learning Intern *Remote*

- Built 5+ ML models across supervised learning and evaluation workflows during a 6-week AI/ML certification, applying the IBM AI toolkit to real-world use cases.
- Applied ML techniques across 8+ real-world datasets, improving model evaluation consistency and reducing error rates in probability-driven use cases.
- Shipped 3 end-to-end ML pipelines – from data ingestion to deployment – for classification and prediction use cases.

Softcrowd Technologies **January 2026 – February 2026**
Data Science Intern *Nashik, India*

- Cleaned and preprocessed 12+ enterprise datasets with Pandas/NumPy, cutting data-prep time by 30%.
- Trained and tuned Random Forest models via cross-validation, reaching 91% prediction accuracy.
- Presented findings to 5+ cross-functional stakeholders, informing data-driven decisions across the full pipeline from framing to delivery.

Projects

Multimodal Deep Learning System for Diabetic Retinopathy Staging with Clinical Deferral Mechanism

- Co-developed a multimodal deep learning pipeline fusing retinal fundus images with patient clinical data (age, diabetes duration, blood pressure) using a fine-tuned CNN (ResNet18 / EfficientNet-B0) and a feed-forward neural network for 5-class disease staging.
- Designed a contrastive-learning-based deferral mechanism to flag low-confidence and low-quality cases for clinician review, improving diagnostic reliability in low-resource settings.
- Integrated model explainability using Grad-CAM and SHAP, and built a Streamlit dashboard for real-time result reporting.

Bharat Autonomous Governance Agent (BAGA)

- Built an LLM-powered governance platform classifying citizen complaints with 92% accuracy using LangChain and the OpenAI API.
- Shipped a Next.js frontend and Flask backend integrating a Random Forest classifier with a Firebase database for real-time grievance routing and tracking.
- Co-led design, development, and the final judges' pitch of the complete system within a 12-hour hackathon window.

Achievements

- Won Special Prize at BLOCKVERSE 2026 Hackathon, GDGC SNJB (April 2026), for building BAGA within a 12-hour window.
- Recognized among the top teams for taking an autonomous governance platform from concept to a live judges' pitch.

Certifications

- AI & Machine Learning Certification — AICTE, IBM & Edunet Foundation (2026).